

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Simulation Task

COURSE NAME: Embedded Systems

COURSE CODE: ECA1421

COURSE OUTCOME COVERED

CO2: Apply Embedded C programming concepts, including structured design, real-time constraint handling, and SystemC-based modeling, to develop and analyze embedded applications for industrial use cases. (BL5)

Name of the Student :

Register Number :

Type Your Question Here:

Requirements Analysis & Conceptual Understanding (20)	System Design & Simulation Model Development(25)	Implementation & Tool Usage (20)	Testing, Validation & Result Analysis (20)	Documentation(15)	Total(100)

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

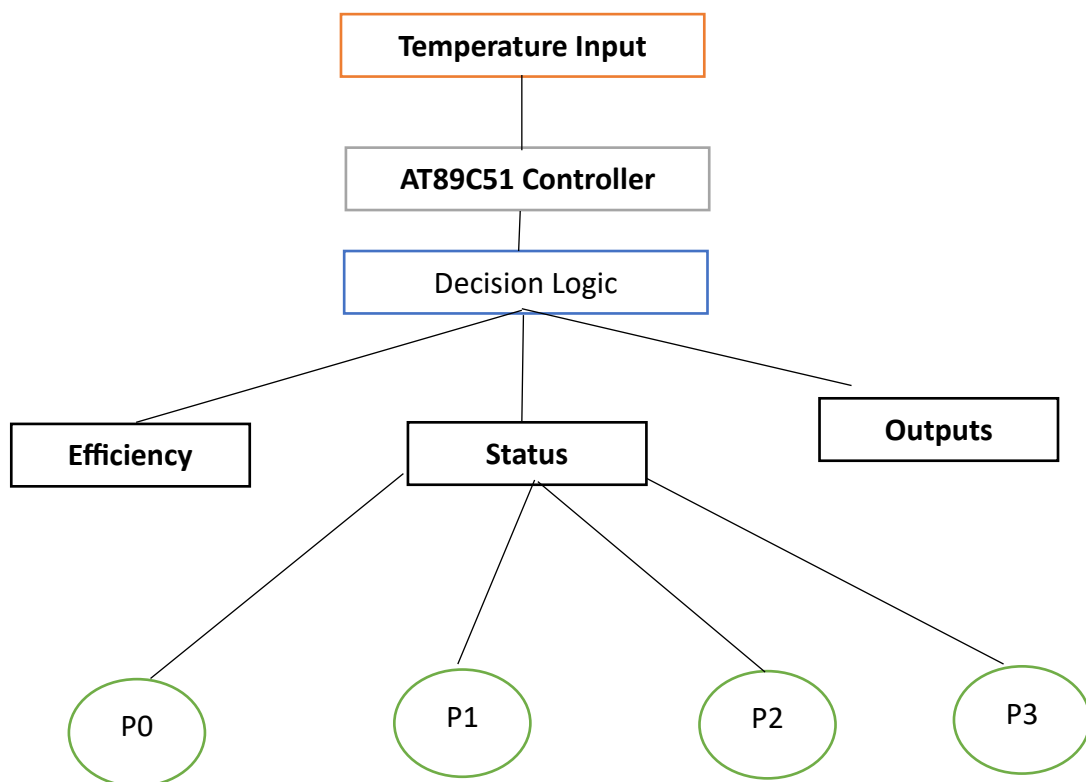
Create a comprehensive Embedded C simulation for an Industry 5.0 smart system with performance evaluation

Aim : Industry 5.0 aims to integrate human intelligence with smart automation to improve industrial productivity, efficiency, and sustainability. Industries require intelligent monitoring systems capable of evaluating machine performance based on operational parameters such as temperature and load.

Objectives

- Monitor machine temperature and load.
- Evaluate system efficiency.
- Determine machine operating status.
- Display performance parameters through microcontroller ports.
- Simulate the system using Embedded C in Keil μ Vision.

System Design & Simulation Model Development



Input Parameter	Value
Temperature	65°C
Load	75%
Threshold Temperature	80°C
Threshold Load	90%

Tool Usage :

Software Tool Keil μ Vision 5

Programming Language Embedded C

Target Device AT89C51

Program

```
#include <reg51.h>

unsigned int temperature = 65;
unsigned int load = 75;
unsigned long efficiency;
unsigned char status;

void main()
{
    if(temperature < 80 && load < 90)
    {
        efficiency = 95;
        status = 0x01;
    }
    else if(temperature >= 80 && load < 90)
    {
        efficiency = 75;
        status = 0x02;
    }
}
```

```
}  
else  
{  
    efficiency = 50;  
    status = 0x03;  
}  
P0 = (unsigned char)efficiency;  
P1 = status;  
P2 = temperature;  
P3 = load;  
while(1);  
}
```

Procedure

1. Open Keil μ Vision.
2. Create a new project and select AT89C51.
3. Create a new C file and enter the program.
4. Save the file and add it to the project.
5. Build the project using F7.
6. Start the Debug Session.
7. Open Watch Window.
8. Add:
 - temperature
 - load
 - efficiency
 - status
9. Run the program using F5.
10. Observe Watch Window and Port outputs.

C:\Users\jadal\OneDrive\Documents\embedded\simulation\task\uproj - pVision

The screenshot shows the uVision IDE interface. The main window displays the assembly code for a simulation task. The code is organized into sections for Parallel Port 0, 1, 2, and 3. The code includes instructions like SJMP, LJMP, and AUIP. The bottom panel shows the 'Registers' window with values for r0 through r7, Sys, a, b, sp_max, ddr, PC, \$, states, sec, and psw. The right panel shows the 'Watch' window with variables like temperature, load, efficiency, and status.

Registers:

Register	Value
r0	0x0e
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x01
r7	0x00
Sys	0x00
a	0x00
b	0x00
sp_max	0x10
ddr	0x08
PC	0x08EE
\$	11192916
states	121.4509
sec	0x00
psw	0x00

Watch:

Name	Value	Type
temperature	0x0041	uint
load	0x0048	uint
efficiency	0x0000005F	ulong
status	0x01	uchar

Code:

```

32: while(1);
C:0x08EE SJMP C:08EE
C:0x08F0 LJMP C:0F00
C:0x08F3 AUIP C:0A02

simulation task
1 #include <reg51.h>
2
3 unsigned int temperature = 65;
4 unsigned int load = 75;
5
6 unsigned long efficiency;
7 unsigned char status;
8
9 void main()
10 {
11     if(temperature < 80 && load < 90)
12     {
13         efficiency = 95;
14         status = 0x01;
15     }
16     else if(temperature >= 80 && load < 90)
17     {
18         efficiency = 75;
19         status = 0x02;
20     }
21     else
22     {
23         efficiency = 50;
24         status = 0x03;
25     }

```

Result

The Industry 5.0 Smart Factory Monitoring and Performance Evaluation System was successfully implemented and simulated using Embedded C in Keil μ Vision. The system effectively monitored temperature and load parameters, evaluated efficiency, determined machine status, and displayed the results through microcontroller ports.